MATH4515 Statistical and Computational Methods in Mathematical Finance
2026-2027 Fall Semester
Assignment 2

Submission deadline of Assignment 2: 11:59p.m. of 3rd Nov, 2026 (Tues)

Instruction: Please complete all required problems. Full details (including description of methods used and explanation, key formula and theorem used and final answer) must be shown **clearly** to receive full credits. Marks can be deducted for incomplete solution or unclear solution. Additional score will be given for well-written assignment.

Please submit your completed work via the submission system in canvas before the deadline. Late assignment will not be accepted.

Your submission must be (1) hand-written (you may write on the ipad if you wish), (2) in a single pdf. file (other file formats will not be accepted) and (3) contain your full name and student ID on the first page of the assignment.

Problem 1 (Required, 30 marks)

Calculate the following quantities using Monte Carlo simulation. Choose the number of simulation runs such that the percentage error is less than 0.5%.

(a) $\mathbb{E}[X^4]$ where X has standard normal distribution.

(b) $\mathbb{E}[\min(\max(X-1000,0),6000)]$ where X has gamma distribution with the probability density function

$$f_X(x) = \frac{\left(\frac{x}{\theta}\right)^{\alpha} e^{-\frac{x}{\theta}}}{x\Gamma(\alpha)}, \qquad x > 0$$

Here, $\alpha = 3.6$ is *shape* parameter and $\theta = 1000$ is scale parameter.
(*Hint: You can check the use of "rgamma" in RStudio by typing "help(rgamma)")

(c) $P(X_1 + X_2 + X_3 > 9)$, where X_1, X_2 and X_3 are independent and identically distributed random variables which X_i has general t-distribution with $\mu = 1$, $\lambda = 1$ and degree of freedom $v = 3.5$.

Explain your calculation.
(*Note: For this problem, please include the full program code in your solution. You may leave your explanation in the program code)

Problem 2 (Required, 35 marks)

For each of the following questions, write a RStudio program for generating the random numbers from the given distribution. Please explain the construction of your algorithm besides the sample R-code. Write your explanation (or relevant calculation) separately.

(a) X is continuous random variable with the probability density function:

$$f_X(x) = x^2 e^{-\frac{x^3}{3}}, \qquad x > 0.$$

(b) N is discrete random variables with the probability mass function:

$$P(N = 0) = 0.6,$$
$$P(N = k) = \frac{C}{k!}, \quad k = 1,2,3, \ldots.$$

where $C > 0$ is normalizing constant.

(😊Hint: Find the value of C first.)

(c) Y is a continuous random variable with the probability density function:

$$f_Y(y) = \frac{C}{1 + y^6}, \qquad y > 0,$$

where C is the normalizing constant.

(😊Hint: You may consider another probability density function defined by $g_Y(y) = \frac{5}{(1+y)^6}, y > 0$)

Problem 3 (Required, 35 marks)

(a) We let X be a random variable which has the following probability density function:

$$f_X(x) = \frac{\alpha}{x^{\alpha+1}}, \qquad x \geq 1.$$

The excel file "MATH4515_HW2_Data1" contains sample data drawn from the distribution of X.

(i) Determine the MLE of α.

(ii) Estimating the mean square error using bootstrapping method. Choose the number of simulation runs to be $B = 10000$.

(b) The excel files "MATH4515_HW2_Data2" contains the sample daily logreturn of a stock. We let μ be the average daily logreturn of the stock. Determine 98% confidence interval of μ using suitable method. Explain your calculation.

Bonus Problem (Optional, 40 marks)

Complete the following tasks using Monte Carlo simulation.

(a) Estimate $\mathbb{E}[\max(4800 - 140X - X^2, 0)]$ where X has the normal distribution with mean 10 and variance 100. Choose the number of simulation runs such that the percentage error is smaller than 0.5%

(b) We X_1, X_2, X_3 are independent and identically distributed which each X_i has the following probability density function:

$$f_X(x) = Ce^{-x^4}, \qquad x \geq 0.$$

where C is the normalizing constant.

We let $Y = X_1 + X_2 + X_3$, obtain the probability distribution of Y. Select a reasonable value of the simulation runs that can visualize the distribution of Y clearly.